Joseph Lee

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Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science, Co-op — *Sep 2024 – Apr 2029*

- President's Scholarship of Distinction
- Relevant Coursework: OOP, Data Structures & Algorithms, Combinatorics, Optimization

Experience

Acceleration Consortium, University of Toronto — Toronto, ON
Software Developer (Full-Stack and Machine Learning) — *Sep 2026 – Apr 2027*

- Built a Next.js and FastAPI dashboard that lets lab scientists predict formulation results and test different inputs without running physical experiments
- Built batch prediction pipelines to replace one-at-a-time runs, reducing prediction costs by ∼95%
- Designed scalable Python REST APIs for spreadsheet data ingestion and validation, standardizing core data processing across new workflows
- Applied Bayesian optimization to select promising experiments based on prior results, reaching target outcomes with 80% fewer experiments than brute-force testing
- Automated AWS deployments with Terraform and GitHub Actions, reducing deployment errors by validating frontend and backend changes before release

Environment and Climate Change Canada — Sept. 2025 – Dec. 2025
Software QA Analyst — *Toronto, ON*

- Engineered automation frameworks in Python, Robot Framework, Playwright, and RPA Framework across 100+ national applications, cutting manual test cycles by 60% and surfacing 20+ defects pre-release
- Built Azure SQL validation pipelines verifying backend data, REST APIs, and large datasets feeding analytics and ML workflows, catching schema drift and data integrity issues pre-production
- Shipped Azure DevOps dashboards tracking defects, performance, and regression coverage, driving defect triage that lowered duplicate report rate by 40% across the team

Environment and Climate Change Canada — May 2025 – Dec. 2025
IT Technician and Support Analyst — *Toronto, ON*

- Delivered Tier 1/2 support for 200+ end users across Windows, Linux, M365, and Outlook server environments, cutting average resolution time by 50% through reusable diagnostic playbooks
- Engineered Python, PowerShell, Power Automate, and SAP GUI scripts and automations to consolidate repetitive cross-system admin tasks, shortening task duration by 80% across daily IT operations

Projects

Quote Finder | *Python, LangChain, FastAPI, Anthropic Claude API, React* | GitHub — 2026

- Built a full-stack RAG application retrieving contextually relevant quotes via vector embeddings and cosine similarity, orchestrating a LangChain and Claude API pipeline behind a FastAPI backend
- Preprocessed and embedded 25000+ quotes into a vector store, tuning chunking and cosine similarity top-k retrieval to improve result relevance and reduce query latency by 50%

Image Lab | *Python, FastAPI, PyTorch, OpenCV, Docker, CustomTkinter* | GitHub — 2025

- Created a multi-interface image processing platform (FastAPI backend, web frontend, desktop GUI) sharing one processing core, with 13+ filters, U2-Net background removal, and ResNet auto-tagging
- Integrated U2-Net for background removal and transfer-learned ResNet/EfficientNet for auto-tagging; containerized with Docker; added CI/CD pipeline with GitHub actions and exposed a documented Python REST API

Technical Skills

Languages: Python, TypeScript/JavaScript, C/C++, Java, SQL
ML/AI: PyTorch, PyTorch Geometric, BoTorch, scikit-learn, Optuna, MLflow, LangChain, SHAP, Pandas, NumPy
Frameworks & Cloud: AWS (ECS, RDS, S3, Terraform, Cognito), Supabase, Docker, Git, FastAPI, Next.js/React, Node.js, Streamlit